

Developer Mock Test Results

Test ID:	e838c9e9-1c58-42...	Student ID:	9333
Test Type:	Developer Assessment	Date:	2026-02-28 16:01:51
Overall Score:	1/25 (4.0%)	Performance:	Needs Improvement
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	0/10	0.0%	Needs Work
MCQ/Theory	0/10	0.0%	Needs Work
Coding	1/5	20.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (0/10 - 0.0%)

Q1. There are 3 switches, but they are not labelled. Each switch corresponds to one of three light bulbs in a room. Each light bulb is either on or off. Y...

Your Answer:

SKIPPED

Correct Answer:

Turn switch 1 to on for 5 minutes, then turn it off. Turn switch 2 to on and enter the room.

The correct answer is A) Turn switch 1 to on for 5 minutes, then turn it off. Turn switch 2 to on and enter the room. This method works because the bulb corresponding to switch 1 will be warm but off, the bulb corresponding to switch 2 will be on, and the bulb corresponding to switch 3 will be off and cold, allowing you to determine which switch corresponds to which bulb. By using the heat of the bulb as an indicator, you can gather the necessary information in a single observation.

Q2. A mixture contains 2 parts of water and 3 parts of sugar. What is the ratio of water to sugar?

Your Answer: SKIPPED

Correct Answer: 2:3

The correct answer is A) 2:3 because the mixture contains 2 parts of water and 3 parts of sugar, directly giving us the ratio of water to sugar. This ratio represents the proportional relationship between the two components in the mixture. The user's skipped answer suggests they may not have understood the direct relationship between the parts of water and sugar given in the question.

Q3. The average of 5 numbers is 20. If one number is 30, what is the sum of the remaining 4 numbers?

Your Answer: SKIPPED

Correct Answer: 60

■ The correct answer is 60 because the average of 5 numbers is 20, so the total sum of the 5 numbers is $5 * 20 = 100$. If one number is 30, then the sum of the remaining 4 numbers is $100 - 30 = 70$. However, the correct calculation yields that the sum of all 5 numbers is 100 and one of the numbers is 30 and the average of 5 is 20, so $5 * 20 = 100$, and $100 - 30$ (one of the 5 numbers) = 70, but since we need the sum of the remaining 4, we know that $100 - 40$ (if the average is 20 and one is 30 then the other 4 must average $(100-30)/4 = 17.5$, so the sum of the 4 must be 70 and one is 30 so the average is 20 or

■ **Q4. What is the next number: 1, 4, 9, 16, 25, ?**

Your Answer: __SKIPPED__

Correct Answer: 36

■ The correct answer is 36 because the sequence consists of square numbers, where each number is the result of multiplying an integer by itself ($1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$, and $6^2 = 36$). The user's answer was skipped, so we can't identify a specific mistake, but the key to solving this sequence is recognizing the pattern of square numbers. By continuing this pattern, we arrive at the correct answer, 36.

■ **Q5. A can do a piece of work in 8 days. B can do the same work in 10 days. C can do the same work in 12 days. How many days will it take for A, B, and C t...**

Your Answer: __SKIPPED__

Correct Answer: 4 days

■ To find the time it takes for A, B, and C to do the work together, we calculate their combined rate of work. A's rate is $1/8$, B's rate is $1/10$, and C's rate is $1/12$, so together they work at a rate of $1/8 + 1/10 + 1/12 = 5/24$ of the work per day. Since their combined rate is $5/24$, it will take them $24/5 = 4.8$ days, which rounds to 4 days when considering whole days for completion, making option B the correct answer.

■ **Q6. A can do a piece of work in 10 days. B can do the same work in 15 days. How many days will it take for A and B to do the work together?**

Your Answer: __SKIPPED__

Correct Answer: 6 days

■ To find the number of days it will take for A and B to do the work together, we need to calculate their combined work rate. A's work rate is $1/10$ of the work per day, and B's work rate is $1/15$ of the work per day, so together they can complete $1/10 + 1/15 = 1/6$ of the work per day. Therefore, it will take them 6 days to complete the work together, making option B the correct answer.

■ **Q7. A bag contains 5 red balls and 3 blue balls. What is the ratio of red balls to blue balls?**

Your Answer: __SKIPPED__

Correct Answer: 5:3

■ The correct answer is C) 5:3 because the bag contains 5 red balls and 3 blue balls, so the ratio of red to blue is directly 5:3. This is a simple ratio problem where the number of each type of ball is compared directly. The user should have counted the number of red and blue balls and expressed them as a ratio, rather than simplifying or altering the numbers in any way.

■ **Q8. A shopkeeper buys an item for Rs.100 and sells it for Rs.120. What is his profit percentage?**

Your Answer: __SKIPPED__

Correct Answer: 20%

■ The correct answer is 20% because the shopkeeper's profit is Rs.20 ($\text{Rs.}120 - \text{Rs.}100$), and to find the profit percentage, we divide the profit by the cost price and multiply by 100, which gives $(20/100) * 100 = 20\%$. The user's answer is not provided, but if they chose a different option, it would be due to a miscalculation of the profit percentage. This calculation is a basic example of finding profit percentage, which is a fundamental concept in business and finance.

■ **Q9. The average of 3 numbers is 15. What is the sum of the 3 numbers?**

Your Answer: __SKIPPED__

Correct Answer: 45

■ The correct answer is 45 because the average of 3 numbers is calculated by dividing the sum of the numbers by 3. Since the average is given as 15, the sum of the 3 numbers can be found by multiplying the average by 3, which gives $15 * 3 = 45$. The user's answer is not provided, but any answer other than 45 would be incorrect, as it would not accurately represent the sum of the 3 numbers with an average of 15.

■ Q10. There are 5 houses in a row, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different n...

Your Answer: __SKIPPED__

Correct Answer: Blue

■ The correct answer is Blue because the Canadian lives in the first house, and since the American lives in the red house, the Canadian cannot live in the red house. Given the other clues, the only colour left for the first house that satisfies all conditions is blue. The user's mistake would be not carefully analyzing the given clues to determine the correct colour of the Canadian's house.

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is the purpose of using locks in Python threading?

Your Answer: __SKIPPED__

Correct Answer: To prevent data corruption and ensure thread safety

■ The correct answer, "To prevent data corruption and ensure thread safety", is right because locks in Python threading are used to synchronize access to shared resources, preventing multiple threads from modifying the same data simultaneously, which can lead to data corruption. This ensures that threads execute safely and reliably, without interfering with each other. By using locks, developers can prevent thread-related issues and ensure the integrity of their program's data.

■ Q2. What is the first step in the thread lifecycle?

Your Answer: __SKIPPED__

Correct Answer: Import the threading module

■ The correct answer is B) Import the threading module, as this is the initial step to work with threads in Python. The user's answer was skipped, but typically, the mistake would be choosing an action that occurs later in the thread lifecycle, such as starting the thread or waiting for its completion. Importing the threading module is the first step, as it allows you to access the necessary classes and functions to create and manage threads.

■ Q3. What is the purpose of the Global Interpreter Lock (GIL) in Python?

Your Answer:

__SKIPPED__

Correct Answer:

To prevent multiple threads from executing Python bytecode concurrently

■ The correct answer is B) To prevent multiple threads from executing Python bytecode concurrently, because the Global Interpreter Lock (GIL) is a mechanism used in Python to synchronize access to Python objects, preventing multiple threads from executing Python bytecodes at once. This is done to prevent issues with memory management and to ensure thread safety. The GIL effectively limits the benefits of multithreading in CPU-bound tasks, but it does provide a simpler and more efficient way to handle thread safety in I/O-bound tasks.

■ Q4. What is the correct step to read data from a file in Python?

Your Answer:

__SKIPPED__

Correct Answer:

Open the file using the open() function, read data using read(), close the file

■ The correct answer, B) Open the file using the open() function, read data using read(), close the file, is the right step to read data from a file in Python. This is because the `open()` function is used to access the file, the `read()` function is used to retrieve the data, and closing the file is important to free up system resources. The other options are incorrect because they involve writing or appending data, or using non-existent functions like `delete()`, which is not a valid method for file operations in Python.

■ Q5. What is the correct order of steps to execute a thread in Python?

Your Answer:

__SKIPPED__

Correct Answer:

```
Import the threading module, define a function to run in a thread, create a Thread object, start the thread
```

■ The correct answer is C) Import the threading module, define a function to run in a thread, create a Thread object, start the thread. This is because to execute a thread in Python, you first need to import the threading module, then define a function that will run in the thread, followed by creating a Thread object with the defined function, and finally starting the thread. The other options are incorrect because they either miss a crucial step or have the steps in the wrong order, such as trying to start a thread before defining the function it will run or creating the Thread object.

■ Q6. What is a best practice for using multithreading in Python?

Your Answer:

__SKIPPED__

Correct Answer:

```
Use multithreading for I/O-bound tasks
```

■ The correct answer is B) Use multithreading for I/O-bound tasks, because Python's Global Interpreter Lock (GIL) prevents multiple threads from executing CPU-bound tasks concurrently, making multithreading less effective for such tasks. However, for I/O-bound tasks like network requests or file operations, multithreading can significantly improve performance by allowing other threads to run while waiting for I/O operations to complete. Using multithreading for I/O-bound tasks can help utilize system resources more efficiently and improve overall program responsiveness.

■ Q7. Why is synchronization required in multithreading?

Your Answer: __SKIPPED__

Correct Answer: To prevent data corruption and ensure data integrity

■ The correct answer is B) To prevent data corruption and ensure data integrity. Synchronization is required in multithreading to prevent multiple threads from accessing and modifying shared data simultaneously, which can lead to data corruption and inconsistencies. By synchronizing threads, we can ensure that only one thread can access and modify the data at a time, thereby maintaining data integrity and preventing errors.

■ Q8. What is the first step in reading and writing files in Python?

Your Answer:

__SKIPPED__

Correct Answer:

```
Open the file using the open() function
```

■ The correct answer is C) Open the file using the open() function because this is the initial step required to access a file in Python. Before you can read or write to a file, you must first open it using the open() function, which returns a file object that can be used for further operations. The other options, such as reading or writing data, can only be performed after the file has been successfully opened.

■ Q9. What is the primary benefit of using multithreading in Python?

Your Answer: __SKIPPED__

Correct Answer: Increased responsiveness and performance in I/O-bound applications

■ The correct answer, "Increased responsiveness and performance in I/O-bound applications", is right because multithreading in Python allows for concurrent execution of threads, which improves the overall responsiveness and performance of applications that spend most of their time waiting for input/output operations to complete. This is particularly useful in applications that involve network requests, database queries, or file operations. The other options are incorrect because multithreading in Python does not necessarily improve CPU performance due to the Global Interpreter Lock (GIL), and it does not directly enhance memory management or simplify error handling.

■ Q10. What is the purpose of using try and except blocks in Python file handling?

Your Answer:

__SKIPPED__

Correct Answer:

To handle errors and exceptions during file operations

■ The correct answer is "To handle errors and exceptions during file operations" because try-except blocks in Python are used to catch and manage exceptions that may occur during file handling, such as file not found or permission errors. This allows the program to gracefully handle errors and prevent crashes, ensuring a more robust and reliable file handling process. The user's skipped answer suggests a lack of understanding of the purpose of try-except blocks in Python file handling, which is a crucial aspect of programming to handle potential errors and exceptions.

■ CODING SECTION (1/5 - 20.0%)

■ Q1. Write a Python program to download multiple files concurrently using threads. ****Input:**** A list of file URLs (e.g., `["http://example.com/file1.txt", ...`

Your Answer:

```
import threading
import requests
import os

def download_file(url):
    try:
        filename = url.split("/")[-1]
        response = requests.get(url)

        with open(filename, "wb") as file:
            file.write(response.content)

        print(f"Downloaded: {filename}")

    except Exception as e:
        print(f"Failed to download {url}: {e}")

# Input list of URLs
urls = [
    "http://example.com/file1.txt",
    "http://example.com/file2.txt",
    "http://example.com/file3.txt"
]

threads = []

# Create and start threads
for url in urls:
    thread = threading.Thread(target=download_file, args=(url,))
    thread.start()
    threads.append(thread)

# Wait for all threads to finish
for thread in threads:
    thread.join()

print("All files downloaded")
```

Correct Answer:

```
import requests
import threading

def download_file(url):
    filename = url.split("/")[-1]
    response = requests.get(url)
    with open(filename, "wb") as file:
```

```

        file.write(response.content)

def main():
    urls = input().strip().split()
    threads = []

    for url in urls:
        thread = threading.Thread(target=download_file, args=(url,))
        thread.start()
        threads.append(thread)

    for thread in threads:
        thread.join()

    print("All files downloaded")

if __name__ == "__main__":
    main()

```

■ The student's code is close to correct, but it's missing the main function to manage the threads and handle the input. The corrected code should use a main function to create and start threads for each URL, and then wait for all threads to finish before printing the success message. Additionally, error handling should be improved to handle potential exceptions during file downloads.

■ **Q2. Write a Python program to count the number of lines in a given text file. ***Input:*** The name of the text file (e.g., "example.txt") ***Output:*** The n...**

Your Answer:

```

# Program to count number of lines in a file

filename = "example.txt"

try:
    with open(filename, "r") as file:
        lines = file.readlines()
        print(len(lines))

except FileNotFoundError:
    print("Error: File not found")
except Exception:
    print("Error: Unable to read file")

```

Correct Answer:

```

def count_lines_in_file(filename):
    try:
        with open(filename, 'r') as file:
            lines = sum(1 for line in file)
            return lines
    except FileNotFoundError:
        return "File not found"

filename = input("Enter the filename: ")
print(count_lines_in_file(filename))

```

■ The student's code is mostly correct, but it doesn't ask for the filename as input and doesn't handle the case where the file is found but can't be read. To fix this, the code should be wrapped in a function that takes the filename as an argument and includes a more specific exception handling, as shown in the corrected code. Additionally, using `sum(1 for line in file)` is more memory-efficient than `file.readlines()` for large files.

■ **Q3. Write a Python function to calculate the average grade of a student given their scores in three subjects: math, science, and English. ***Input:*** A dic...**

Your Answer:

```
def calculate_average(grades):
    total = sum(grades.values())
    average = total / len(grades)
    return round(average, 2)

# Example input
student_scores = {"math": 85, "science": 90, "english": 78}

# Output
print(calculate_average(student_scores))
```

■ Your code is well-structured and effectively solves the problem, passing all tests with ease. Great job on writing clean, efficient, and accurate code!

■ **Q4. Write a Python class to manage bank account transactions. The class should have methods to deposit, withdraw, and check the balance. **Input:** Initia...**

Your Answer:

```
class BankAccount:
    def __init__(self, balance):
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
        else:
            print("Insufficient funds")

    def get_balance(self):
        return self.balance

# Input
initial_balance = 100
transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)]

# Create account
account = BankAccount(initial_balance)

# Process transactions
for action, amount in transactions:
    if action == "deposit":
        account.deposit(amount)
    elif action == "withdraw":
        account.withdraw(amount)

# Output final balance
print(account.get_balance())
```

Correct Answer:

```
class BankAccount:
    def __init__(self, balance=0):
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
```

```

def withdraw(self, amount):
    if amount > self.balance:
        print("Insufficient balance")
    else:
        self.balance -= amount

def check_balance(self):
    return self.balance

def main():
    initial_balance = int(input())
    transactions = input()[1:-1].split(",")
    transactions = [tuple(transaction.split(", ")) for transaction in transactions]
    transactions = [(transaction[0][1:-1], int(transaction[1])) for transaction in transactions]

    account = BankAccount(initial_balance)
    for transaction in transactions:
        if transaction[0] == "deposit":
            account.deposit(transaction[1])
        elif transaction[0] == "withdraw":
            account.withdraw(transaction[1])

    print(account.check_balance())

if __name__ == "__main__":
    main()

```

■ The student's code is incomplete and lacks a method to process a list of transactions. To fix this, a new method should be added to iterate over the transactions and call the corresponding methods (`deposit` or `withdraw`). Additionally, a `main` function should be implemented to create a `BankAccount` object, process the transactions, and print the final balance. Here is the corrected code: ``python class BankAccount: def __init__(self, balance=0): self.balance = balance def deposit(self, amount): self.balance += amount def withdraw(self, amount): if amount > self.balance: print("Insufficient balance") else: self.balance -= amount def check_balance(self): return self.balance def main(): initial_balance = 100 transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)] account = BankAccount(initial_balance) for transaction

■ **Q5. Write a Python program to crawl multiple web pages concurrently using threads. ****Input:**** A list of URLs (e.g., `["http://example.com/page1", "http://...**

Your Answer:

```

import threading
import requests

def crawl_page(url):
    try:
        response = requests.get(url, timeout=10)
        print(f"Crawled: {url} | Status Code: {response.status_code}")
    except Exception as e:
        print(f"Failed to crawl {url}: {e}")

# Input list of URLs
urls = [
    "http://example.com/page1",
    "http://example.com/page2",
    "http://example.com/page3"
]

threads = []

# Create and start threads

```

```

for url in urls:
    thread = threading.Thread(target=crawl_page, args=(url,))
    thread.start()
    threads.append(thread)

# Wait for all threads to finish
for thread in threads:
    thread.join()

print("All pages crawled")

```

Correct Answer:

```

import requests
import threading

def crawl_page(url):
    try:
        response = requests.get(url)
        print(f"Crawled {url} with status code {response.status_code}")
    except Exception as e:
        print(f"Error crawling {url}: {str(e)}")

def main():
    urls = input().split()
    threads = []

    for url in urls:
        thread = threading.Thread(target=crawl_page, args=(url,))
        threads.append(thread)
        thread.start()

    for thread in threads:
        thread.join()

    print("All pages crawled")

if __name__ == "__main__":
    main()

```

■ The issue with the student's code is that it doesn't actually create and start the threads to crawl the pages concurrently. To fix this, the student should create a `threading.Thread` object for each URL, set its target to the `crawl_page` function, and then start the thread. Additionally, the student should use a `main` function to encapsulate the program's entry point and handle the input URLs. Here is the corrected code snippet: ``python import requests import threading def crawl_page(url): try: response = requests.get(url) print(f"Crawled {url} with status code {response.status_code}") except Exception as e: print(f"Error crawling {url}: {str(e)}") def main(): urls = input().split() threads = [] for url in urls: thread = threading.Thread(target=crawl_page, args=(url,)) thread.start() threads.append(thread) for thread

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.